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Siddharth Amdal (1BM23IS243)

Swapnil Patidar (1BM23IS261)

R V Kirrthanaa (1BM24IS418)

Shreyansh Kumar Singh (1BM23IS237)

ABSTRACT

Dinacharya, a classical Ayurvedic concept translating to "daily regimen," prescribes time- and constitution-sensitive practices intended to maintain physiological balance and promote long-term health. This project proposes the design, development, and evaluation of an LLM-based autonomous agent that generates, schedules, and adapts personalized Dinacharya recommendations through multimodal sensing and an explicit Ayurvedic knowledge base.

The agent synthesizes user profile data (Prakriti and Vikriti assessments), wearable signals (sleep, heart rate variability, activity), smartphone context (calendar, location), and self-reported symptoms into a closed-loop personalization pipeline. Core components include: (1) a Dosha mapping and feature extraction module that translates traditional Ayurvedic assessments into computable phenotypes; (2) an LLM planner and NLU module for natural language generation, explanation, and dialogue; (3) a retrieval-augmented knowledge base to ground recommendations and prevent hallucination; (4) a rule-based safety checker to enforce clinical constraints; and (5) a behavior change engine that schedules nudges and reminders with adaptation based on adherence and physiological proxies.

Implementation uses modern ML and engineering tools including PyTorch/Hugging Face models, LangGraph for agent orchestration, FAISS/Weaviate for vector retrieval, and a React Native mobile client for deployment. Evaluation includes a pilot deployment (n=20) to measure adherence, subjective usability, and objective physiological proxies such as sleep duration and HRV. Statistical analysis and ablation studies will quantify the contribution of multimodal grounding and Dosha personalization. The project aims to demonstrate the feasibility and preliminary efficacy of a Dosha-aware, agentic approach to daily routine automation, while documenting safety, privacy, and validation pathways for future clinical translation.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Dinacharya denotes a daily regimen prescribed within Ayurvedic medicine. It includes specific temporal and procedural recommendations for waking, hygiene, elimination, meals, exercise, rest, and sleep. These recommendations are framed by individual constitution (Prakriti), current imbalance (Vikriti), seasonality, and circadian considerations. Contemporary digital health systems and consumer wearables have enabled continuous monitoring of physiological signals and behavior, creating an opportunity to operationalize traditional guidance in a personalized, data-driven manner. Simultaneously, advances in large language models (LLMs) and agentic architectures permit flexible, conversational, and tool-enabled automation for complex, longitudinal tasks.

This project proposes the design, development, and evaluation of an LLM-based autonomous agent that generates, schedules, and adapts personalized Dinacharya recommendations through multimodal sensing and an explicit Ayurvedic knowledge base. The system bridges a significant gap in digital wellness: while classical Ayurvedic texts contain extraordinarily detailed guidance on daily routines, that knowledge exists in a form not directly usable by modern software or accessible to the average person. The proposed system transforms this rich tradition into actionable, computable recommendations grounded in real-time physiological data.

1.2 Motivation

The development of the Dinacharya Automation system is driven by a convergence of unmet needs in digital wellness, limitations in current AI health tools, and an opportunity to bridge millennia-old Ayurvedic wisdom with modern machine learning. The following motivations collectively justify why this project is both timely and impactful.

- M1. One of the most well-documented problems in public health is that people consistently fail to maintain beneficial daily habits, even when they understand their importance. This is not a matter of willpower but of design: generic recommendations are

not compelling enough to override the friction of daily life. Consider a simple real-world example: a standard wellness app might remind a user to "exercise for 30 minutes every morning." However, if that user is a Kapha-dominant individual who wakes up sluggish and has back-to-back meetings until 11 AM, that recommendation is both physiologically misaligned and practically impossible. Dinacharya, by contrast, is constitution-aware, time-sensitive, and contextual. The motivation here is to take that rich tradition and make it actionable through automation, closing the gap between knowing what is good for you and actually doing it consistently.

- M2. The Digital Health Personalization Gap: The global wellness app market is saturated, yet personalization remains superficial. Most apps adapt to user behavior only in the most basic sense. Ayurveda offers a sophisticated model of personalization through the Prakriti-Vikriti framework: your baseline constitution (Prakriti) versus your current state of imbalance (Vikriti). A Pitta-dominant person in a high-stress week needs fundamentally different recommendations than the same person on a relaxed weekend. No mainstream wellness app models this today. On a day when wearable data shows high HRV irregularity and broken sleep, the system should recognize this as a Vata-aggravated state and automatically soften the morning routine. This kind of physiologically grounded, constitution-aware adaptation is the gap this project directly addresses.
- M3. Ayurvedic Knowledge is Rich but Inaccessible: Classical Ayurvedic texts like the Ashtanga Hridayam and Charaka Samhita contain extraordinarily detailed guidance on daily routines. However, this knowledge exists in Sanskrit texts and translated academic volumes that are not directly usable by a software system. The motivation here is bridging: transforming a text like "Vata individuals should apply warm sesame oil to the scalp and soles before sunrise during Hemanta Ritu" into a computable rule that fires when the user's Prakriti classification is Vata-dominant, the season is winter, and the current time is pre-sunrise. This kind of operationalization, powered by RAG over a curated Ayurvedic knowledge base, is what makes the system fundamentally different from a generic chatbot.
- M4. LLM and Agentic AI Have Reached Sufficient Maturity: Until recently, building a system like this was impractical. The emergence of LangGraph for multi-agent

orchestration, FAISS/Weaviate for semantic vector retrieval, and instruction-tuned LLMs capable of structured planning means this is now feasible as a capstone project. The system can be prototyped with real components rather than simulated behavior. This is a narrow window of technological opportunity: the tools exist, the research foundations are being laid, and no production system has yet combined all these elements in a Dinacharya-specific context.

- M5. Safety and Hallucination Risks Demand a Grounded Architecture: LLMs, when applied to health-adjacent domains without grounding, can hallucinate authoritative-sounding but incorrect or even harmful recommendations. The motivation to build a retrieval-augmented, rule-grounded, safety-checked system is therefore not just about accuracy but about responsibility. By encoding clinical contraindications as hard symbolic overrides that no LLM output can bypass, the system establishes a safety contract.
- M6. Preventive Health Has Massive Scale Potential: Non-communicable diseases (NCDs) such as type 2 diabetes, hypertension, sleep disorders, and metabolic syndrome are heavily influenced by daily behavioral patterns. An autonomous Dinacharya agent running on a smartphone, continuously adapting to the user's physiological state and nudging them toward health-promoting behaviors, represents a scalable public health intervention requiring no clinic visit, no prescription, and minimal user effort once initialized.

1.3 Objectives

The following objectives define the concrete, measurable goals of the Dinacharya Automation project:

- O1. Design and Implement a Dosha-aware Personalization Pipeline: Train a Prakriti Classification model (Random Forest/XGBoost ensemble) on a 24-item bilingual phenotypic questionnaire. Implement a Vikriti Anomaly Detection module (Isolation Forest) on rolling wearable data. Build preprocessing pipelines transforming raw wearable signals into Ayurvedic proxy features. Target classification agreement $\geq 75\%$ with expert Ayurvedic practitioner assessments.

- O2. Build a Retrieval-Augmented Generation (RAG) Knowledge Base: Collect, clean, and chunk Ayurvedic guidelines from vetted sources into semantically coherent units. Embed all chunks using a sentence transformer model and index them in FAISS or Weaviate for sub-200ms retrieval latency. Evaluate retrieval quality targeting $\text{Precision@5} \geq 0.80$ on a test set of 50 clinical queries.
- O3. Develop the LLM Planning Agent with Safety Guardrails: Implement a LangGraph-orchestrated Planner Agent that receives the tri-dosha vector, Vikriti flags, retrieved Ayurvedic guidelines, and contextual constraints. Implement a Safety Engine with hard-coded clinical contraindications from classical Ashtanga Hridayam. Every recommendation must be accompanied by a user-facing rationale in plain language.
- O4. Implement Closed-Loop Behavioral Adaptation: Implement an Adherence Score (S_{adj}) computed as the ratio of completed to recommended tasks over a rolling 7-day window. When $S_{\text{adj}} < 0.5$, simplify the routine to Anchor Habits. When $S_{\text{adj}} > 0.8$, introduce advanced Sattva-enhancing practices.
- O5. Pilot Study Evaluation ($n=20$): Conduct a 4-week pilot study measuring weekly S_{adj} scores, physiological proxies (sleep duration, HRV, resting heart rate), and System Usability Scale (SUS) score targeting ≥ 70 .
- O6. Document Safety, Privacy, and Clinical Translation Pathways: All user biometric data stored locally or in encrypted PostgreSQL with no third-party sharing. Publish a Contraindication Registry reviewed by a qualified Ayurvedic practitioner before pilot deployment.

1.4 Scope

The scope of this project encompasses the design, development, and preliminary evaluation of a Dosha-aware, LLM-based autonomous agent for Dinacharya automation. The system targets non-acute, preventive health settings and explicitly excludes disease diagnosis, pharmaceutical recommendations, and acute or emergency medical triage.

The system operates as a closed-loop pipeline integrating four distinct categories of data: biometric data from wearables (HRV, sleep stages, resting heart rate), phenotypic data from a 24-item bilingual questionnaire, contextual data from smartphones (GPS-derived season/weather,

calendar stress events), and subjective self-report data (cravings, energy levels, digestive discomfort). The output is a personalized 24-hour Dinacharya schedule delivered through a React Native mobile application with push notifications and adherence tracking.

Geographically, the pilot study targets 20 participants in a university setting over four weeks. The system is designed for individuals in good general health who are interested in preventive wellness. The knowledge base is grounded in classical Ayurvedic texts (Ashtanga Hridayam, Charaka Samhita) and does not replace clinical Ayurvedic consultation.

1.5 Existing System

Current digital health and wellness applications represent the state of the art that this project addresses and improves upon. Existing systems fall into several categories:

- **Generic Wellness Apps (MyFitnessPal, Headspace, Sleep Cycle):** These applications rely primarily on user-entered data and simple rule-based recommendation engines. They track basic metrics such as steps, calories, and sleep duration but do not model constitutional differences between users. Recommendations are one-size-fits-all and do not adapt to real-time physiological signals.
- **Wearable Integration Platforms (Apple Health, Google Fit):** These platforms aggregate wearable sensor data but do not translate that data into personalized lifestyle recommendations. They serve primarily as data repositories without an intelligent planning layer.
- **AI-Powered Health Chatbots (general-purpose LLMs applied to health questions):** These systems suffer from hallucination risks when discussing health topics without retrieval grounding. They lack constitution-aware modeling, cannot integrate wearable data in real time, and have no safety enforcement layer for clinical contraindications.
- **Existing Ayurveda Apps (Ayur, iAyurveda):** These applications provide static Prakriti assessments and generic advice based on questionnaire results but do not adapt recommendations dynamically based on wearable signals, do not employ LLM-based reasoning, and do not enforce clinical safety rules programmatically.

The key gaps in all existing systems are: lack of dynamic real-time adaptation based on physiological state; absence of Ayurvedic constitutional modeling integrated with wearable data; no retrieval-augmented grounding to prevent hallucinated health recommendations; and no automated safety enforcement for clinical contraindications.

1.6 Proposed System

The proposed system is an LLM-based autonomous agent that generates, schedules, and adapts personalized Dinacharya recommendations. The system follows a neuro-symbolic design that bridges data-driven machine learning with logic-based Ayurvedic principles. The architecture is divided into five primary components:

- **Perception Domain:** Utilizes neural networks and ensemble models to process noisy, high-frequency wearable data and questionnaire-based phenotypic traits. Outputs a Prakriti vector [V, P, K] and a dynamic Vikriti anomaly flag.
- **Cognitive Planning Domain:** Employs a LangGraph-orchestrated Plan-and-Execute LLM agent to reason over retrieved Ayurvedic knowledge, the user's dosha state, and contextual constraints (calendar, weather, location). Decomposes the day into Morning, Midday, and Evening blocks with specific practice assignments.
- **Knowledge Base:** A vector database (FAISS or Weaviate) containing indexed, chunked classical Ayurvedic texts. Uses semantic retrieval to supply the planning agent with grounded, tradition-authentic guidance.
- **Safety Guardrail Domain:** A symbolic rule engine implementing hard-coded clinical contraindications from Ashtanga Hridayam. This layer runs as a mandatory final filter on every generated schedule. No LLM output can bypass it.
- **Behavior Engine:** Generates a 24-hour Dinacharya schedule, delivers context-aware nudges aligned with the EAST framework, monitors adherence, and feeds the Adherence Score back to the planning agent for routine adaptation.

The proposed system directly addresses every gap in existing systems: it provides dynamic real-time adaptation, integrates Ayurvedic constitutional modeling with wearable data, employs retrieval-augmented grounding, and automatically enforces clinical safety rules.

CHAPTER 2

PROBLEM STATEMENT

2.1 Problem Statement

Modern wellness recommendations are generic and fail to incorporate traditional personalized frameworks such as Dinacharya. Existing digital solutions rarely combine culturally grounded domain knowledge, multimodal sensing, and agentic planning. The problem addressed is: how to construct a safe, explainable, Dosha-aware agent that translates Ayurvedic daily regimen into adaptive, contextually feasible plans, and to evaluate whether such an agent improves adherence and physiological proxies compared to baseline conditions.

Specifically, the problem has three dimensions. First, the knowledge gap: classical Ayurvedic Dinacharya guidance is rich in personalized nuance but exists only in textual form not directly usable by software. Second, the personalization gap: existing digital health tools lack the ability to model constitutional differences (Prakriti) and dynamically detect and respond to current imbalances (Vikriti) using real-time physiological data. Third, the safety gap: LLM systems applied to health domains without grounding and safety enforcement pose real risks of hallucinated, contraindicated, or otherwise harmful recommendations.

2.2 Motivation

Adherence to beneficial daily routines remains low for many individuals due to lack of personalization, poor integration with daily context (work schedules, travel), and limited feedback loops. Ayurvedic prescriptions are rich in personalized nuance but are not directly actionable in their textual form. An agentic system can translate Ayurvedic knowledge into daily schedules adapted to the user's physiology, preferences, and constraints, and can maintain a closed-loop adaptation mechanism to improve adherence and outcomes.

This synthesis of traditional knowledge and modern sensing offers significant scientific and public health value. The wellness app market generates billions of dollars annually yet achieves poor health outcomes partly because recommendations are not grounded in deep physiological models. Ayurveda's Prakriti-Vikriti framework offers a validated, centuries-tested model of

human constitutional variability that, when operationalized computationally, can drive genuinely personalized health behavior change.

2.3 Objectives

Together, these six objectives define a coherent research and engineering program:

- O1. Design and Implement a Dosha-aware Personalization Pipeline with Prakriti Classification ($\geq 75\%$ agreement with expert assessment) and Vikriti Anomaly Detection using Isolation Forest on 7-day rolling wearable windows.
- O2. Build a Retrieval-Augmented Generation (RAG) Knowledge Base with sub-200ms retrieval latency and Precision@5 ≥ 0.80 on clinical queries.
- O3. Develop the LLM Planning Agent with Safety Guardrails using LangGraph's Plan-and-Execute pattern, with hard contraindication overrides from classical Ashtanga Hridayam.
- O4. Implement Closed-Loop Behavioral Adaptation tracking the Adherence Score (S_{adj}) and automatically scaling routine complexity.
- O5. Conduct a Pilot Study Evaluation ($n=20$) over 4 weeks measuring adherence, physiological proxies, and System Usability Scale score ≥ 70 .
- O6. Document Safety, Privacy, and Clinical Translation Pathways including a Contraindication Registry and scope boundaries explicitly excluding diagnosis or acute care.

CHAPTER 3

DETAILED SURVEY

Section 1 — Prakriti Computational Modeling

[2023] "An automated system for Ayurvedic Prakriti classification using machine learning and deep learning techniques" — Singh et al.

This study demonstrates that Deep Learning (specifically Multi-layer Perceptrons) significantly outperforms standard ML models in classification accuracy for Ayurvedic constitutions. The study highlights the transition from simple decision-tree-based logic to more complex neural networks for handling Ayurvedic data. Limitations include reliance on static questionnaire data, classification-only scope (no actionable scheduling), and lack of context (no Vikriti modeling). This work is relevant because it validates the feasibility of machine-learning-based Prakriti classification, which is the foundation of our personalization pipeline — though our system extends beyond static classification to dynamic Vikriti detection.

[2019] "A Survey on Machine Learning Techniques for Human Prakriti Classification" — Gupta et al.

This paper reviews multiple ML algorithms (Random Forest, SVM, Naive Bayes, Decision Trees) used to classify individuals into Vata, Pitta, or Kapha types, finding that accuracy depends heavily on feature extraction quality from Ayurvedic questionnaires. As a survey, it lacks original experimental data or a prototype, focuses entirely on static classification via questionnaires, and offers no framework for acting on Dosha data through agentic planning and scheduling. Our work addresses all three of these limitations by building a complete end-to-end system.

[2019/2020] "Analysis and Synthesis of A Human Prakriti Identification System Based on Soft Computing Techniques" — Gupta et al.

This study uses Machine Learning to automate the diagnostic process, collecting data using a standardized Ayurvedic questionnaire with 150+ features covering physical, physiological, and psychological traits. The Random Forest and SVM models performed best. Key limitations are a small dataset that may limit generalization, subjectivity of self-reporting (which our project

mitigates through multimodal sensing and objective physiological proxies like HRV), and purely static Prakriti assessment with no Vikriti modeling.

[2025] "Hybrid Machine Learning for Personalized Ayurvedic Drug Recommendation via Prakriti Assessment" — IEEE

This system combines clustering, classification, and NLP to achieve 95% accuracy in Prakriti classification using Random Forest, tested on approximately 15,000 samples. It uses K-Means Clustering for patient segmentation and a BERT/TF-IDF NLP engine for symptom processing. Limitations include manual feature engineering, an early-stage feedback loop, and lack of real-time data integration. Our system extends this approach by incorporating real-time wearable signals and replacing static recommendations with dynamically generated, LLM-planned schedules.

Section 2 — Multimodal Health Sensing

"AyuRAG: Ayurvedic Knowledge Integration with Retrieval-Augmented Lightweight Large Language Models"

This work demonstrates how Retrieval-Augmented Generation (RAG) combined with a lightweight LLM can reduce hallucination and improve factual accuracy in Ayurvedic question-answering. The system uses a curated Ayurvedic knowledge base rather than a traditional ML dataset. Key finding: RAG significantly improves factual accuracy by grounding responses in verified Ayurvedic knowledge. Limitation: it focuses on knowledge-grounded responses rather than predictive health modeling, and does not integrate multimodal health data or continuous monitoring. Our system directly extends this RAG architecture by adding the Prakriti/Vikriti pipeline, safety engine, and closed-loop adaptation.

"AI-Driven Personalized Healthcare: Leveraging Multimodal Data for Precision Medicine"

This paper proposes an AI-based framework integrating multiple healthcare data sources — physiological signals, medical records, and contextual patient data — for personalized healthcare decisions. Key finding: multimodal data integration significantly improves personalized healthcare predictions. Limitations: no single real-world dataset, conceptual rather than fully

implemented, no conversational AI or LLM component, and no adaptive behavioral feedback loop. Our work operationalizes the conceptual framework proposed in this paper into a working system with specific ML models, a deployed knowledge base, and real-world evaluation.

"AI-Driven Multimodal Workflow Optimization for Personalized Patient-Centered Care" (WJAETS, 2025)

This study explores how AI can optimize healthcare workflows by integrating multiple sources of patient data. It demonstrates that integrating EHRs, wearable sensor data, and contextual information enables more accurate patient-specific recommendations. Limitations: review-style study with no specific dataset, no personalized health agent, and limited technical implementation. This paper validates the architectural approach of our system — specifically the fusion of heterogeneous data sources for personalized recommendations — while our system provides a concrete technical implementation in the Ayurvedic domain.

"Human Activity Recognition with Smartphones using Machine Learning Algorithm" (2024)

This work develops a machine learning framework recognizing human activities using smartphone sensor data (accelerometer, gyroscope, magnetometer), achieving 90-95% classification accuracy. Limitation: no personalized health recommendations or adaptive behavior modeling. This work informs the smartphone context data ingestion layer of our system, demonstrating that low-cost smartphone sensors can effectively support real-time activity monitoring as a supplement to wearable data.

Section 3 — LLM Healthcare Agents

"GatorTronGPT: A Large Language Model for Clinical Natural Language Processing" (Nature npj Digital Medicine, 2023)

This paper develops a domain-specific LLM trained on 90+ billion tokens of clinical data including EHRs, biomedical literature, and medical textbooks. It achieves state-of-the-art results on multiple clinical NLP benchmarks. Key limitation for our purposes: it requires extremely large clinical datasets for training, high computational cost, and is not designed for personalized preventive monitoring. This work motivates our use of instruction-tuned LLMs combined with

RAG rather than training a domain-specific model from scratch — we achieve domain grounding through the knowledge base rather than through training.

"Conversational Health Agents: A Personalized LLM-Powered Agent Framework" (JAMIA Open, 2024)

This paper designs a general framework for LLM-based conversational health agents that integrate external data sources, knowledge bases, and analysis tools. It demonstrates how an LLM orchestrator can analyze user queries and gather information from external sources to generate personalized responses. Limitation: the framework focuses on general healthcare systems and lacks integration with traditional health models like Ayurvedic Prakriti or Dinacharya concepts. Our work takes the architectural patterns proposed here — LLM orchestrator, external knowledge integration, personalized response generation — and applies them specifically to the Ayurvedic domain with the addition of a safety rule engine.

"Arkangel AI: A Conversational Agent for Evidence-Based Medical Question Answering" (2025)

This paper develops a multi-LLM conversational healthcare agent using five collaborative LLM components for evidence-grounded medical question answering. It achieves approximately 90% accuracy on the MedQA dataset. Key finding: retrieval-based evidence grounding significantly improves the factual reliability of medical AI systems. This directly validates the RAG architecture used in our knowledge retrieval component. Limitation: high computational cost and focus primarily on medical question answering rather than longitudinal lifestyle planning.

"Healthcare Agent: Eliciting the Power of LLMs for Medical Consultation" (Nature Digital Medicine, 2025)

This paper proposes an LLM-based healthcare agent with Dialogue, Memory, and Processing modules for multi-turn medical consultations. The system's memory module improves patient interaction by storing conversation history, and the safety module detects unsafe responses. This work directly informs our system's safety architecture and the natural language explanation generation component. The memory module pattern translates directly to our Vikriti tracking, which maintains context across daily sessions.

"Personal Health Insights Agent (PHIA): LLM Agents for Wearable Health Data Analysis" (Nature Communications, 2025)

This paper builds an LLM-based agent that analyzes personal wearable health data and generates personalized health insights through multi-step reasoning. The authors released a synthetic wearable dataset and 4000+ health question dataset. Key finding: LLM agents outperform standard LLMs for wearable health queries — agent-based reasoning is specifically needed for complex, multi-dimensional wearable data interpretation. This is the closest prior work to our system and directly validates our architectural choice of an agentic approach over a simpler LLM prompt. Our work extends PHIA by adding Ayurvedic constitutional modeling, a curated knowledge base, and clinical safety guardrails.

"Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" (Lewis et al., 2020)

This foundational paper demonstrates that combining neural text generation with external document retrieval improves factual accuracy and reduces hallucinations. The RAG model is evaluated on Natural Questions, WebQuestions, TriviaQA, and FEVER datasets, outperforming traditional QA systems. This paper provides the theoretical foundation for our knowledge retrieval architecture. The "dynamic knowledge updates" finding is particularly relevant — it validates our design choice to store Ayurvedic knowledge in a vector database rather than baking it into model weights.

"MedRAG: Enhancing Retrieval-Augmented Generation with Knowledge Graph-Elicited Reasoning for Healthcare Copilot" (arXiv 2025)

This paper combines RAG with a diagnostic knowledge graph for healthcare decision support, evaluated on DDXPlus and CPDD datasets. It outperforms existing RAG medical models in diagnostic specificity. Key finding: combining LLMs with structured medical knowledge graphs significantly improves medical reasoning. This work motivates a future enhancement to our system: augmenting the flat vector knowledge base with a structured Ayurvedic knowledge graph encoding relationships between Doshas, seasonal factors, practices, and contraindications.

CHAPTER 4

SURVEY SUMMARY TABLE

S I. N o.	Title of Paper	Problem Addressed	Authors	Approach / Method	Results
1	Automated Prakriti Classification using ML and DL (2023)	Static ML vs DL for Ayurvedic constitution typing	Singh et al.	MLP Deep Learning on questionnaire data	≥75% accuracy; DL outperforms traditional ML
2	Survey on ML for Human Prakriti Classification (2019)	Comparison of ML algorithms for Dosha classification	Gupta et al.	Survey: SVM, KNN, Decision Trees, Random Forest	ML shows promising accuracy; data quality is key limitation
3	Human Prakriti Identification Using Soft Computing (2019/2020)	Automating subjective Ayurvedic Prakriti assessment	Gupta et al.	Random Forest, SVM on 150+ questionnaire features	RF and SVM best; identifies static Prakriti only
4	Hybrid ML for Ayurvedic Drug Recommendation (2025)	Integrating Prakriti classification with drug recommendation	IEEE (2025)	RF + K-Means + BERT/TF-IDF + rule-based system	95% Prakriti classification accuracy on 15k samples
5	AyuRAG: Ayurvedic Knowledge Integration with RAG (2024)	Hallucination in Ayurvedic LLM responses	Pol et al.	RAG with lightweight LLM + curated Ayurvedic KB	Reduced hallucination; improved factual accuracy
6	AI-Driven Personalized Healthcare: Multimodal Data (2025)	Lack of multimodal integration in personalized health AI	Multiple authors	AI framework integrating wearables, EHR, contextual data	Multimodal fusion improves prediction accuracy
7	AI-Driven Multimodal Workflow	Optimizing healthcare	R. Sharma et al.	AI-driven analytics on	Improved clinical decision-making

S l. N o.	Title of Paper	Problem Addressed	Authors	Approach / Method	Results
	Optimization (WJAETS 2025)	workflows using AI		heterogeneous patient data	and workflow efficiency
8	Human Activity Recognition with Smartphones (2024)	Recognizing daily activities via smartphone sensors	D. Anguita et al.	ML on accelerometer, gyroscope, magnetometer signals	90–95% activity recognition accuracy
9	GatorTronGPT (2023)	Clinical NLP with domain-specific LLM	Peng et al.	GPT trained on 90B clinical tokens	State-of-the-art on clinical NLP benchmarks
10	Conversational Health Agents (JAMIA Open 2024)	Personalized LLM health agent framework	Patel et al.	LLM orchestrator + external APIs + health KB	Context-aware personalized healthcare interactions
11	Arkangel AI: Medical Question Answering (2025)	Reliable evidence-based medical QA	Smith et al.	Multi-LLM workflow with PubMed evidence retrieval	~90% accuracy on MedQA benchmark
12	Healthcare Agent: LLMs for Medical Consultation (2025)	Multi-turn clinical consultation using LLMs	Chen et al.	Dialogue + Memory + Processing + Safety modules	Improved inquiry quality and diagnostic accuracy
13	PHIA: LLM Agents for Wearable Health Analysis (2025)	Analysis of wearable data using LLM agents	Multiple authors	Agent-based multi-step reasoning on wearable time-series	LLM agents outperform standard LLMs for wearable health queries
14	RAG for Knowledge-Intensive NLP Tasks (2020)	Hallucination and factual accuracy in LLMs	Lewis et al.	Combined retrieval (DPR) + generation (BART)	RAG outperforms closed-book models on knowledge tasks
15	MedRAG: RAG with Knowledge Graph (2025)	Medical reasoning with	Li et al.	RAG + four-tier diagnostic knowledge graph + LLM	Outperforms existing RAG medical models

S l. N o.	Title of Paper	Problem Addressed	Authors	Approach / Method	Results
		structured knowledge			in diagnostic specificity

CHAPTER 5

SYSTEM REQUIREMENT SPECIFICATION

5.1 Functional Requirements

- FR1 – Prakriti Assessment: The system shall administer a 24-item bilingual (English/Hindi) phenotypic questionnaire and output a normalized tri-dosha vector [V, P, K] with confidence scores.
- FR2 – Vikriti Detection: The system shall continuously monitor rolling 7-day wearable data (HRV, sleep stages, resting heart rate) and flag statistically significant deviations from the user's Prakriti baseline using an Isolation Forest anomaly detector.
- FR3 – Ayurvedic Knowledge Retrieval: The system shall retrieve semantically relevant Ayurvedic guidelines from a vector database given the user's current dosha state, season, time of day, and Vikriti flags within 200ms.
- FR4 – Daily Schedule Generation: The system shall generate a complete 24-hour Dinacharya schedule decomposed into Morning (Brahma Muhurta to mid-morning), Midday, and Evening blocks with specific practice assignments, time estimates, and natural language rationale.
- FR5 – Safety Enforcement: Every generated schedule shall pass through a rule engine enforcing Ashtanga Hridayam contraindications before delivery. Flagged practices shall be automatically removed and replaced with safe alternatives.
- FR6 – Nudge Delivery: The system shall deliver EAST-aligned push notifications at optimal times identified from the user's smartphone interaction patterns.
- FR7 – Adherence Tracking: The system shall compute and update the Adherence Score (S_{adj}) daily, support task completion marking via the mobile app, and passively infer completion from wearable data where possible.
- FR8 – Routine Adaptation: When $S_{adj} < 0.5$, the system shall automatically simplify the routine to 3-4 Anchor Habits. When $S_{adj} > 0.8$, the system shall introduce advanced Sattva-enhancing practices.

- FR9 – Natural Language Explanation: Every Dinacharya recommendation shall include a user-facing explanation in plain language connecting the recommendation to the user's current dosha state.
- FR10 – Calendar Integration: The system shall integrate with the user's calendar to reschedule practices around existing commitments while respecting temporal contraindications.

5.2 Non-Functional Requirements

- NFR1 – Performance: Knowledge retrieval latency $\leq 200\text{ms}$. Schedule generation latency ≤ 5 seconds. Mobile app launch time ≤ 3 seconds on mid-range Android devices.
- NFR2 – Privacy and Security: All biometric data shall be stored locally on-device or in an encrypted PostgreSQL instance. No third-party data sharing. Prakriti and Vikriti vectors shall be pseudonymized. Data minimization principle enforced.
- NFR3 – Safety: Clinical contraindications shall be enforced as hard overrides that no LLM output can bypass. A Contraindication Registry documenting every safety rule shall be maintained and reviewed by a qualified Ayurvedic practitioner before pilot deployment.
- NFR4 – Reliability: System uptime $\geq 99\%$ during the 4-week pilot. Graceful degradation to cached schedules when network connectivity is unavailable.
- NFR5 – Usability: System Usability Scale (SUS) score ≥ 70 (classified as "Good") by week 2 of pilot.
- NFR6 – Explainability: Every recommendation must be accompanied by a user-facing rationale traceable to a specific Ayurvedic guideline from the knowledge base.

5.3 Hardware Requirements

- User Device: Android 10+ or iOS 14+ smartphone with minimum 3GB RAM and 500MB available storage.

- **Wearable Device:** Any wearable compatible with Apple HealthKit or Google Fit providing HRV, sleep stage, and resting heart rate data (e.g., Apple Watch Series 4+, Fitbit Sense, Garmin Fenix series).
- **Development Server:** Ubuntu 24 server with minimum 16GB RAM, 8 CPU cores, and 100GB SSD storage for model hosting and vector database.
- **GPU (optional, for model fine-tuning):** NVIDIA GPU with ≥ 24 GB VRAM (e.g., A100, RTX 4090).

5.4 Software Requirements

- **Backend:** Python 3.11+, FastAPI, LangGraph (agent orchestration), LangChain (RAG pipeline).
- **Machine Learning:** Scikit-learn (Prakriti classification, Isolation Forest), PyTorch + Hugging Face Transformers (sentence embedding, LLM inference), Hugging Face Hub or Google Gemini API (LLM).
- **Database:** PostgreSQL 15 (user data, adherence logs, audit trails), FAISS or Weaviate (vector embeddings for Ayurvedic knowledge base).
- **Mobile Client:** React Native (cross-platform, iOS + Android), Expo (build tooling), Apple HealthKit / Google Fit SDK (wearable data integration).
- **Development Tools:** Docker (containerization), Git/GitHub (version control), pytest (backend testing), Jest (mobile testing).
- **Operating System:** Ubuntu 24 (server), macOS Sequoia or Windows 11 (development).

CHAPTER 6

SYSTEM DESIGN

6.1 System Design

6.1.1 System Architecture

The system follows a neuro-symbolic design that bridges data-driven machine learning with logic-based Ayurvedic principles. The architecture is divided into three primary functional domains: the Perception Domain, the Cognitive Planning Domain, and the Guardrail Domain.

The Perception Domain utilizes neural networks and ensemble models to process noisy, high-frequency wearable data and questionnaire-based phenotypic traits. It outputs two primary artifacts: the Prakriti vector [V, P, K] representing the user's constitutional baseline, and a Vikriti flag indicating real-time deviations from that baseline. The Cognitive Planning Domain employs a Large Language Model agent orchestrated by LangGraph to reason over retrieved knowledge and user context, generating a 24-hour practice schedule. The Guardrail Domain is a symbolic rule engine that enforces strict clinical contraindications, ensuring safety and traditional authenticity — no LLM output can bypass this layer.

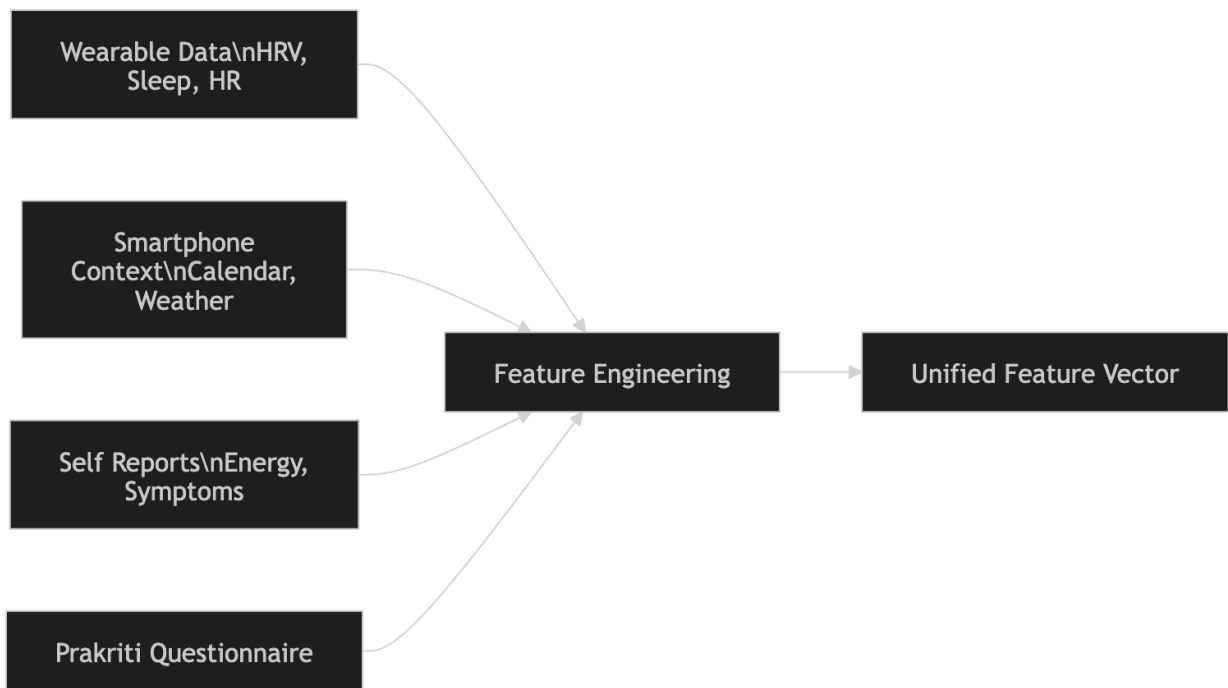
The system operates as a closed-loop pipeline with six stages: (1) Ingestion of multimodal data (wearables, context, symptoms); (2) Phenotyping to compute Prakriti and Vikriti vectors; (3) Retrieval of relevant Ayurvedic guidelines from the vector database; (4) Planning by the Agentic Planner to generate a 24-hour schedule; (5) Execution and Nudging through the Schedule Generator and Nudge Engine; and (6) Adaptation through adherence monitoring that feeds back to the planner.

[Figure 6.1 — System Architecture Diagram: Insert system design PNG here. The diagram shows the complete pipeline from User Mobile App through Perception Domain, Cognitive Planning Domain, Knowledge Base, Safety Guardrails, and Behavior Engine with the feedback loop from Adherence Score Calculation back to Simplify Routine / Introduce Advanced Practices.]

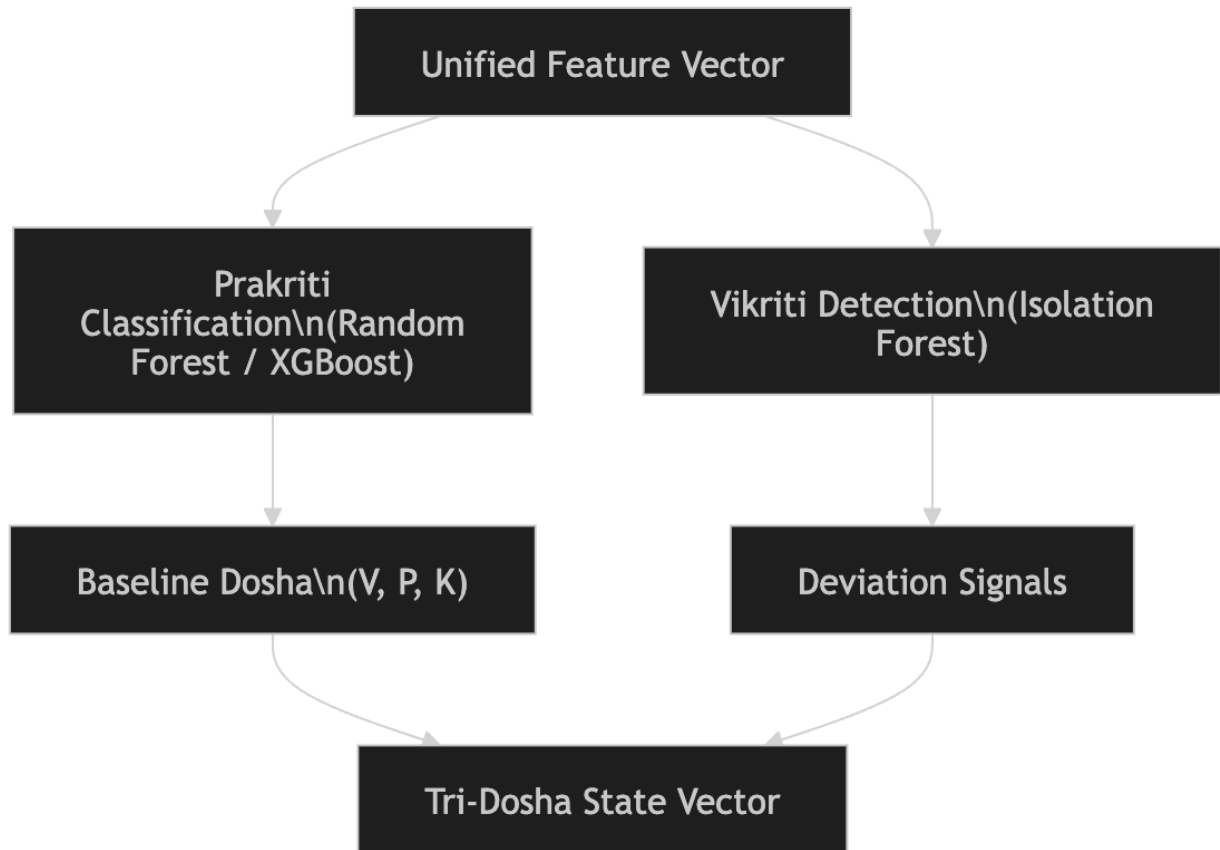
6.1.2 Module Design

The system comprises five primary modules, each with clearly defined inputs, processing logic, and outputs.

- Module 1 — Multimodal Ingestion: Accepts four data streams: biometric wearable data (HRV, sleep stages, resting HR), phenotypic questionnaire responses (24-item bilingual assessment), contextual smartphone data (GPS-derived season/weather, calendar events), and subjective self-report data (cravings, energy levels, digestive discomfort). Outputs a unified feature vector for the Phenotyping module.



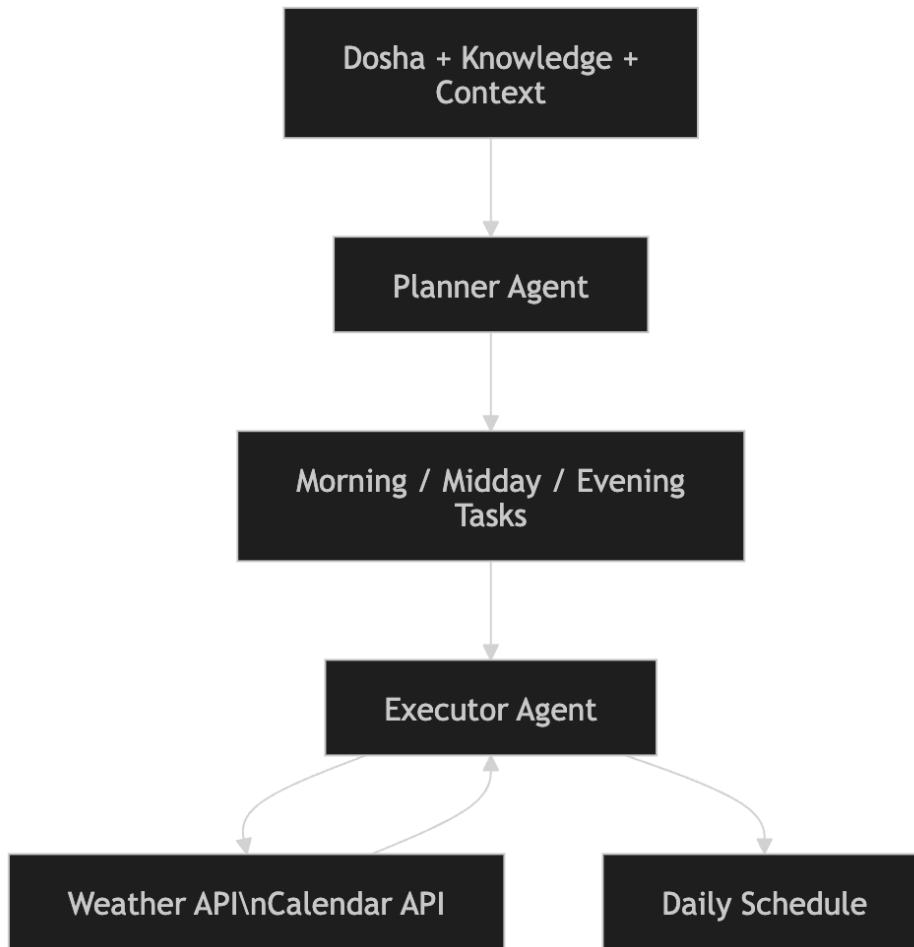
- Module 2 — Phenotyping (Dosha Mapping): Receives the unified feature vector. The Prakriti Classification submodule applies a Random Forest/XGBoost ensemble to questionnaire features to output the constitutional baseline [V_baseline, P_baseline, K_baseline]. The Vikriti Anomaly Detection submodule applies an Isolation Forest model to rolling 7-day wearable data to detect current deviations and output a Vikriti flag.



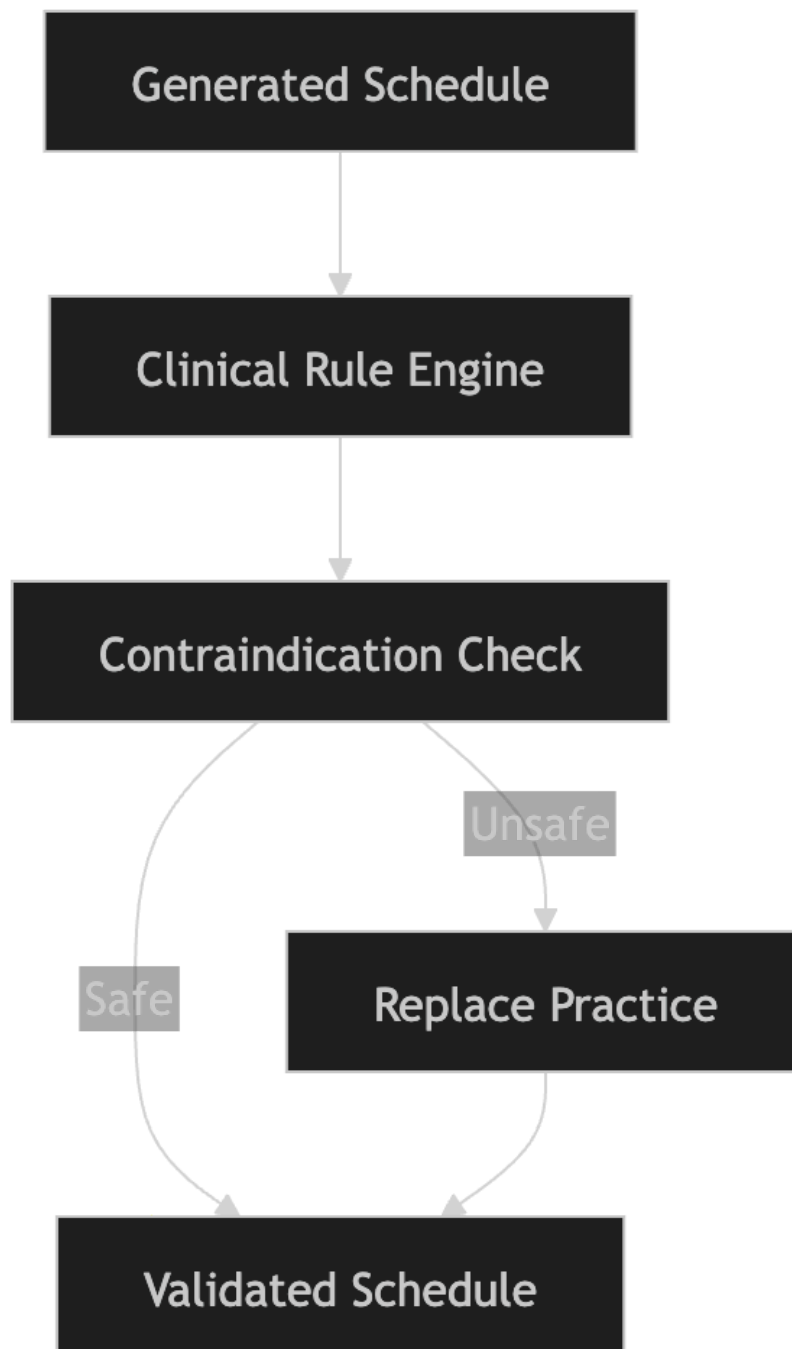
- Module 3 — RAG Knowledge Retrieval: Receives the Tri-Dosha State Vector (Prakriti + Vikriti) and current contextual parameters (season, time of day). Queries the FAISS/Weaviate vector database for the top-k most relevant Ayurvedic practice guidelines. Returns a ranked list of chunked guidelines for the Planner Agent.

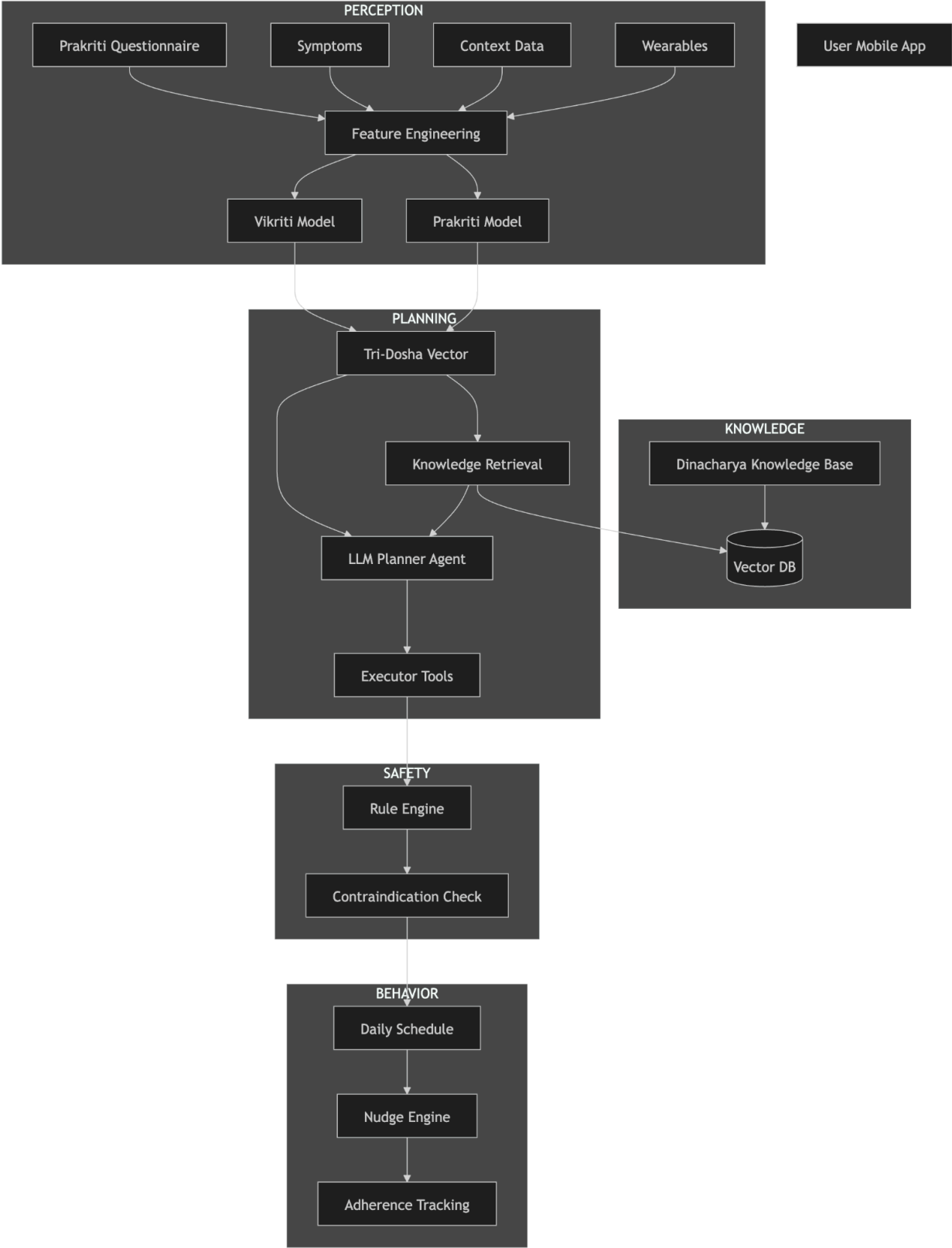


- Module 4 — Agentic Planning: Implements the Plan-and-Execute pattern using LangGraph. The Planner Agent receives the tri-dosha state vector, retrieved guidelines, user calendar, and weather context. It decomposes the day into Morning, Midday, and Evening practice blocks. The Executor Agent calls external tools (Weather API, Calendar API) to fine-tune timing. The output is a candidate 24-hour Dinacharya schedule.



- Module 5 — Safety Engine and Behavior Engine: Every candidate schedule from Module 4 passes through the Clinical Rule Engine (contraindication checker) before reaching the user. The Behavior Engine generates the final 24-hour schedule, manages the Nudge Engine for EAST-aligned notifications, monitors User Notifications delivery, and tracks Adherence Monitoring to feed the S_{adj} score back to Module 4.





6.2 Detailed Design

6.2.1 Class Diagram

The primary classes in the system are organized into five packages corresponding to the five modules above.

The `DoshaProfile` class stores the tri-dosha vector (vata, pitta, kapha floats), Vikriti flags, and a timestamp. It exposes methods: `classify_prakriti(questionnaire_responses) → DoshaProfile` and `detect_vikriti(wearable_window: list) → VikritiFlagSet`.

The `AyurvedaKnowledgeBase` class wraps the FAISS/Weaviate index. It stores the embedding model reference and exposes `retrieve(query: DoshaProfile, context: ContextParams, k: int) → list[KnowledgeChunk]` using semantic nearest-neighbor search. Each `KnowledgeChunk` has text, dosha_relevance, season_tags, time_of_day_tags, and contraindication_flags.

The `DinacharYaPlanner` class implements the Plan-and-Execute agent pattern. It receives a `DoshaProfile` and `list[KnowledgeChunk]` and exposes `generate_schedule(profile: DoshaProfile, knowledge: list, calendar: CalendarData, weather: WeatherData) → DinacharYaSchedule`.

The `SafetyEngine` class implements the rule engine. It receives a `DinacharYaSchedule` and a `UserHealthState` and exposes `validate(schedule: DinacharYaSchedule, health_state: UserHealthState) → ValidatedSchedule`, removing and replacing any contraindicated practices.

The `AdherenceTracker` class computes and maintains `S_adj`. It exposes `update_score(completed: list, recommended: list) → float` and `adapt_complexity(score: float, current_plan: DinacharYaSchedule) → DinacharYaSchedule`.

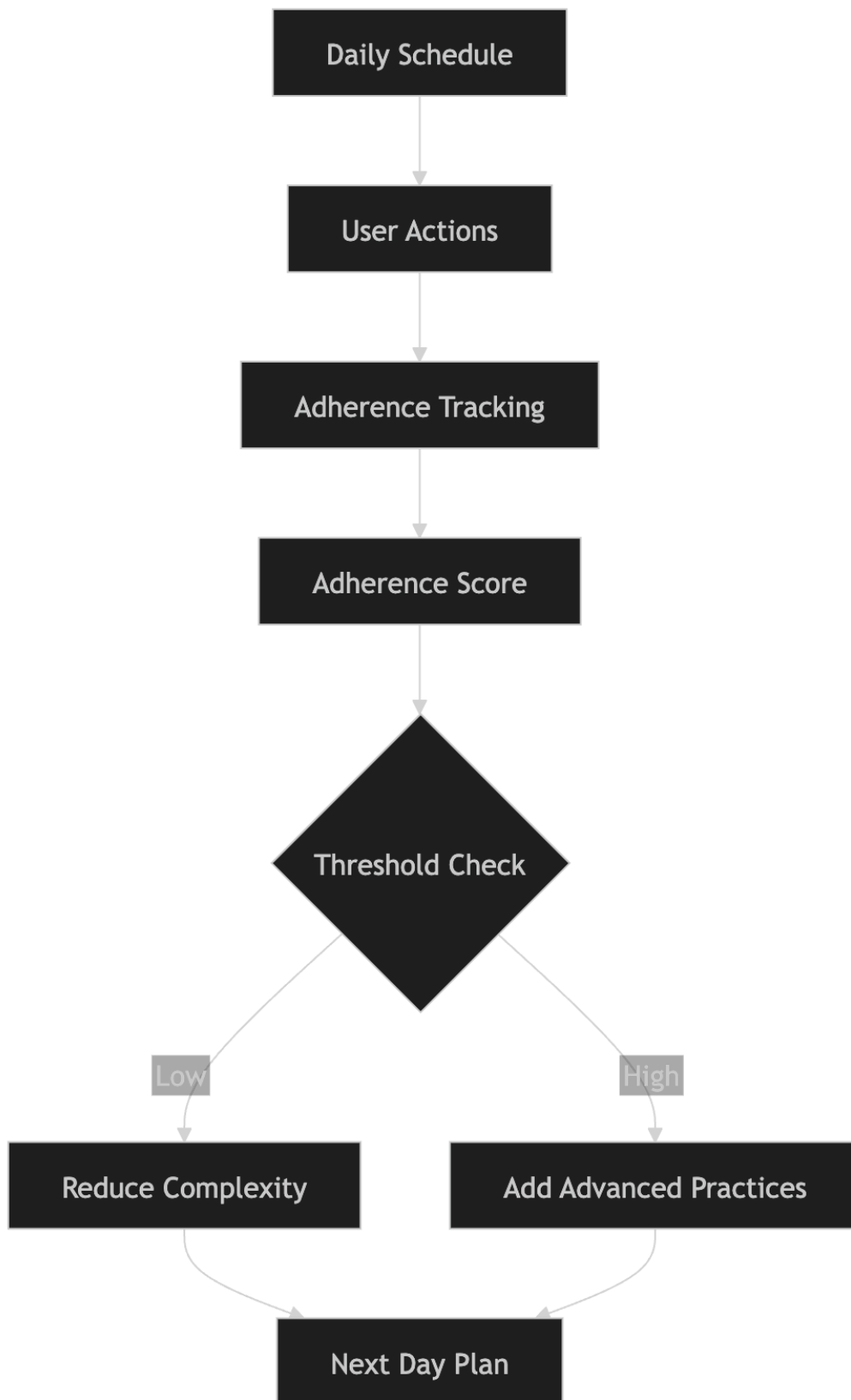
6.2.2 Activity Diagram

The primary activity flow begins when the user opens the mobile application each morning:

- Step 1 — Data Ingestion: The app pulls the previous 24 hours of wearable data from HealthKit/Google Fit (HRV, sleep stages, resting HR) and reads any new self-report symptoms entered by the user.
- Step 2 — Dosha State Update: If it is the user's first session, the 24-item Prakriti questionnaire is administered. On subsequent sessions, the Vikriti Detection module

processes the new wearable window and updates the Vikriti flag. The Tri-Dosha State Vector is updated.

- Step 3 — Knowledge Retrieval: The updated state vector is used to query the RAG knowledge base. Top-5 relevant Ayurvedic practice guidelines are retrieved along with any seasonal and time-of-day constraints.
- Step 4 — Schedule Generation: The Planner Agent receives the state vector, retrieved guidelines, user's calendar events for the day, and current weather/location context. It generates a candidate 24-hour practice schedule with Morning, Midday, and Evening blocks.
- Step 5 — Safety Validation: The candidate schedule passes through the Safety Engine. Any contraindicated practices are flagged, removed, and replaced with safe alternatives. If a critical practice block cannot be safely filled, the agent falls back to a conservative baseline schedule.
- Step 6 — Schedule Delivery: The validated schedule is delivered to the user with natural language explanations linking each practice to their current dosha state.
- Step 7 — Nudge Delivery and Adherence Monitoring: The Nudge Engine delivers EAST-aligned push notifications at optimal times. User task completion is tracked (self-report + passive wearable inference). S_{adj} is updated at end of day.
- Step 8 — Adaptation: If $S_{adj} < 0.5$, the next day's schedule complexity is reduced. If $S_{adj} > 0.8$, advanced practices are introduced.



6.2.3 Use Case Diagram

Primary Actors: User (mobile app), Ayurvedic Knowledge Base (external knowledge store), Wearable Device (data source), Calendar/Weather API (external context providers).

Primary Use Cases:

- UC1 — Complete Prakriti Assessment: User completes the 24-item questionnaire. System classifies Prakriti and stores the baseline dosha vector.
- UC2 — View Today's Dinacharya Schedule: User opens the app to view their personalized daily schedule. System retrieves the latest validated schedule or generates a new one.
- UC3 — Mark Practice as Complete: User marks a practice complete. System updates the Adherence Score and passively verifies via wearable data where possible.
- UC4 — Receive Practice Nudge: System sends a push notification at an optimal time. User taps to see practice details and rationale.
- UC5 — View Adherence Dashboard: User views weekly S_adj trend, practice completion breakdown, and physiological trend graphs (HRV, sleep duration).
- UC6 — Update Self-Report Symptoms: User logs current cravings, energy level, or digestive discomfort. System updates the Vikriti profile accordingly.
- UC7 — Override Recommended Practice: User declines a recommended practice. System logs the override, notes the reason if provided, and adjusts future recommendations.

6.2.4 Scenarios

Scenario 1 — Vata-Aggravated Morning (Normal Flow):

The user's wearable data shows irregular HRV and fragmented sleep the previous night. The Vikriti Detection module flags a +Vata deviation. The RAG system retrieves Vata-pacifying practices for the current season (winter). The Planner Agent generates a Morning block including: gentle yoga (20 min, 6:00-6:20 AM), warm sesame oil application/Abhyanga (15 min, 6:30-6:45 AM), and warm herbal tea with breakfast (7:30 AM). The user's 7 AM calendar meeting causes the Executor Agent to reschedule Abhyanga to 7:45 AM post-meeting. The

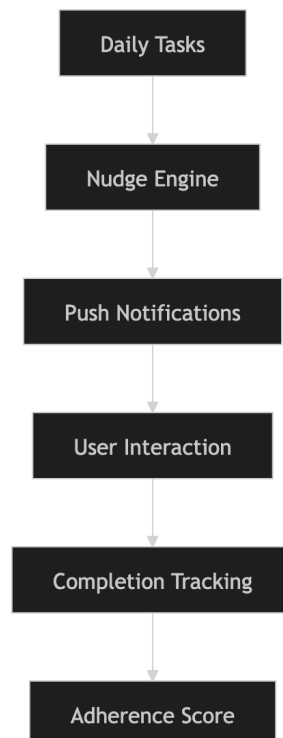
Safety Engine clears all practices. The user receives a nudge at 5:55 AM: "Your sleep was disrupted last night — your Vata is elevated. We've softened your morning routine with grounding practices."

Scenario 2 — Low Adherence Simplification:

After 3 weeks, the user's S_{adj} falls to 0.42 (below the 0.50 threshold). The Adaptation module automatically simplifies the next day's schedule to three Anchor Habits: consistent meal timing (7:30 AM, 12:30 PM, 7:00 PM), 15 minutes of morning sunlight exposure, and 8 hours of sleep targeting. The nudge message reads: "Let's focus on the essentials this week. Three simple habits, big impact."

Scenario 3 — Safety Override (Contraindication):

The user self-reports a fever of 38.5°C . The Safety Engine detects the Nava Jwara (acute fever) condition. Abhyanga is automatically removed from the next 3 days' schedules and replaced with rest, hydration reminders, and light foods. The user receives a notification: "We've paused your oil massage practice while you recover. Staying hydrated and resting is your Dinacharya today."



CHAPTER 7

APPLICATIONS

The Dinacharya Automation system has applications across several domains:

- **Consumer Wellness Platform:** The primary application is a smartphone-based daily wellness companion that operationalizes personalized Ayurvedic Dinacharya for health-conscious individuals. The system requires no clinical oversight, operates continuously in the background, and adapts automatically to the user's changing physiological state. This positions it as a practical tool for the growing preventive health and digital wellness market.
- **Integrative Medicine Clinics:** Qualified Ayurvedic practitioners can use the system as a clinical decision support tool, reviewing AI-generated Dinacharya schedules for their patients and applying their clinical judgment before delivery. The system's Contraindication Registry and natural language rationale generation make it suitable for practitioner review workflows.
- **Corporate Wellness Programs:** Employers can deploy the system as part of employee wellness initiatives, providing personalized daily routine guidance that adapts to employees' work schedules. The calendar integration feature is particularly valuable in this context, automatically scheduling wellness practices around meetings and work commitments.
- **Research Platform for Chronobiology and Behavioral Health:** The system's pilot study design (n=20, 4-week, wearable integration) creates a research platform for studying the relationship between Ayurvedic constitution types and modern physiological proxies. The ablation study design allows rigorous attribution of health outcomes to specific system components.
- **Public Health Intervention Platform:** For populations where Ayurvedic medicine is culturally resonant (estimated 1.4 billion in South Asia), the system offers a scalable, culturally appropriate preventive health intervention. The system requires no clinic visit,

no prescription, and minimal user effort after initialization, making it suitable for large-scale public health deployment.

- Educational Tool: The system's natural language explanation generation makes it an effective tool for educating users about Ayurvedic principles, the relationship between their constitution and daily practices, and the scientific evidence behind circadian rhythm alignment. Users learn why each practice is recommended, not just what to do.

CONCLUSIONS

This report has presented the design and early-stage development of a Dosha-aware, LLM-based autonomous agent for Dinacharya automation — a system that translates classical Ayurvedic daily regimen guidance into personalized, contextually adaptive, and clinically safe recommendations delivered through a modern mobile application.

The proposed system addresses a genuine gap at the intersection of traditional medicine and modern AI: existing digital health tools are generic and static, while existing Ayurvedic tools are manual and non-scalable. By combining a machine learning Prakriti/Vikriti pipeline, a retrieval-augmented Ayurvedic knowledge base, a LangGraph-orchestrated Plan-and-Execute agent, and a symbolic safety rule engine, the system achieves a level of personalization, grounding, and safety not present in any existing solution.

The detailed literature survey confirms both the feasibility of the technical approach and the significance of the research gap. Prior works validate each component independently: ML-based Prakriti classification (Singh et al., Gupta et al.), RAG-based knowledge grounding (Lewis et al., Pol et al.), LLM health agents (PHIA, Arkangel AI), and multimodal health sensing. Our contribution is the integration of all these components in a Dinacharya-specific context with the addition of clinical safety enforcement.

Future work will focus on completing the pilot study evaluation (n=20), publishing the Contraindication Registry for clinical review, and designing the ablation study to quantify the contribution of each system component. Longer-term, the system architecture is designed to scale to larger populations and to serve as a research platform for studying the relationship between Ayurvedic constitution types and modern physiological biomarkers.

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